

HW 7

1. a) Prove the product of two orthogonal matrices is orthogonal.

Let $A, B \in \mathbb{R}^{n \times n}$ be orthogonal matrices. We must prove that $AB(AB)^T = I$.

$$AB(AB)^T = AB \underbrace{B^T A^T}_{I \text{ since } B^{-1} = B^T} = A I A^T = A A^T = I$$

therefore, the product of two orthogonal matrices is orthogonal.

b) If $Q_1 \in \mathbb{R}^{n \times n}$ and $Q_2 \in \mathbb{R}^{n \times n}$ are orthogonal, then $Q = \begin{pmatrix} Q_1 & 0 \\ 0 & Q_2 \end{pmatrix}$ is also orthogonal.

$$Q^T = \begin{pmatrix} Q_1 & 0 \\ 0 & Q_2 \end{pmatrix}^T = \begin{pmatrix} Q_1^T & 0 \\ 0 & Q_2^T \end{pmatrix}$$

$$QQ^T = \begin{pmatrix} Q_1 & 0 \\ 0 & Q_2 \end{pmatrix} \begin{pmatrix} Q_1^T & 0 \\ 0 & Q_2^T \end{pmatrix} = \begin{pmatrix} Q_1 Q_1^T & 0 \\ 0 & Q_2 Q_2^T \end{pmatrix} = \begin{pmatrix} I_{n \times n} & 0 \\ 0 & I_{n \times n} \end{pmatrix} = I_{(n+n) \times (n+n)}$$

therefore, the block matrix Q is also orthogonal.

2. Compute a QR factorization of $A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & -2 & 0 \\ 1 & -2 & 2 \end{pmatrix}$ using Householder reflectors.

$$H_1: y = v_1 - \|v_1\|_2 e_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} \sqrt{2} \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 1-\sqrt{2} \\ 0 \\ 1 \end{bmatrix}$$

$$u = \frac{y}{\|y\|_2} = \frac{1}{\sqrt{2\sqrt{2}+4}} \begin{bmatrix} 1-\sqrt{2} \\ 0 \\ 1 \end{bmatrix}$$

$$\hookrightarrow (1-\sqrt{2})^2 + 1^2 = 1 - 2\sqrt{2} + 2 + 1 = 2\sqrt{2} + 4$$

$$\frac{3+\sqrt{2}(2-\sqrt{2})}{2+\sqrt{2}(2-\sqrt{2})} = \frac{6-3\sqrt{2}+4\sqrt{2}-4}{4-2} = 1 + \frac{1}{\sqrt{2}}$$

$$Q_1 = I - 2uu^T = I - \frac{2}{\|u\|_2} uv^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \frac{2}{4+2\sqrt{2}} \begin{bmatrix} 1+\sqrt{2} \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} 1+\sqrt{2} & 0 & 1 \end{bmatrix} = I - \frac{1}{2+\sqrt{2}} \begin{bmatrix} 3+2\sqrt{2} & 0 & 1+\sqrt{2} \\ 0 & 0 & 0 \\ 1+\sqrt{2} & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{1}{2+\sqrt{2}} & 0 & -\frac{1}{2+\sqrt{2}} \\ 0 & 1 & 0 \\ -\frac{1}{2+\sqrt{2}} & 0 & \frac{1}{2+\sqrt{2}} \end{bmatrix}$$

$$\frac{2+\sqrt{2}-1-\sqrt{2}}{2+\sqrt{2}} = \frac{1}{2+\sqrt{2}} \quad \frac{1+\sqrt{2}}{2+\sqrt{2}}$$

$$H_1 A = \begin{bmatrix} -\frac{1}{2+\sqrt{2}} & 0 & -\frac{1}{2+\sqrt{2}} \\ 0 & 1 & 0 \\ -\frac{1}{2+\sqrt{2}} & 0 & \frac{1}{2+\sqrt{2}} \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & -2 & 0 \\ 1 & -2 & 2 \end{bmatrix} = \begin{bmatrix} -\sqrt{2} & \sqrt{2} & -\frac{3}{\sqrt{2}} \\ 0 & -2 & 0 \\ 0 & -\sqrt{2} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\text{submatrix } x = \begin{bmatrix} -2 & 0 \\ -\sqrt{2} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\|x_2\| = \sqrt{4-2} = \sqrt{2}$$

$$v_2 = x_2 - \|x_2\| e_1 = \begin{bmatrix} 2-\sqrt{2} \\ -\sqrt{2} \end{bmatrix}$$

$$H'_2 = I - \frac{2}{\|v_2\|} v_2 v_2^T = \begin{bmatrix} -\frac{2}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & -\frac{2}{\sqrt{2}} \end{bmatrix}$$

$$H_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -\frac{2}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ 0 & -\frac{1}{\sqrt{2}} & -\frac{2}{\sqrt{2}} \end{bmatrix}$$

$$R = H_2(H_1 A) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -\frac{2}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ 0 & -\frac{1}{\sqrt{2}} & -\frac{2}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} -\sqrt{2} & \sqrt{2} & -\frac{3}{\sqrt{2}} \\ 0 & -2 & 0 \\ 0 & -\sqrt{2} & \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} -\sqrt{2} & \sqrt{2} & -\frac{3}{\sqrt{2}} \\ 0 & \sqrt{2} & -\frac{1}{\sqrt{2}} \\ 0 & 0 & \frac{3}{\sqrt{2}} \end{bmatrix}$$

$$Q = H_1 H_2 = \begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

R

My HW 6 Gram-Schmidt QR result had positive signs on the diagonal of R which results when we normalise vectors. Householder has some negative diagonal entries, though both factorizations work.

3. Let $\|\cdot\|$ be a matrix norm on $\mathbb{R}^{n \times n}$

a) Show that $\|\cdot\|$ is submultiplicative, or for any $A, B \in \mathbb{R}^{n \times n}$, $\|AB\| \leq \|A\| \|B\|$.

$$\|(AB)x\| \leq \|A\| \|Bx\| = \|A\| \|B\| \|x\| \Rightarrow \|(AB)x\| \leq (\|A\| \|B\|) \|x\|$$

$$\hookrightarrow \|Bx\| \leq \|B\| \|x\| \quad (\text{using } \|Ax\| \leq \|A\| \|x\|)$$

Then, using the property $\|A\| = \max_{v \neq 0} \frac{\|Av\|}{\|v\|}$,

$$\|(AB)x\| \leq (\|A\| \|B\|) \|x\| \Rightarrow \frac{\|(AB)x\|}{\|x\|} \leq \|A\| \|B\| \Rightarrow \max_{x \neq 0} \frac{\|(AB)x\|}{\|x\|} \leq \|A\| \|B\|$$

since $\|A\| \|B\|$ is an upper bound

for $\frac{\|(AB)x\|}{\|x\|}$ for all x , it is

also $\geq$ the supremum (max).

$$\Rightarrow \|AB\| \leq \|A\| \|B\| \quad \checkmark$$

b) Show that $\|AX - I\| \leq \kappa(A) \|XA - I\|$

Use $I = AA^{-1}$ and $AX = AX(AX)^{-1}$

$$AX - I = AX - AA^{-1} = A(XA - I)A^{-1}$$

$$\|A(XA - I)A^{-1}\| \leq \|A\| \|XA - I\| \|A^{-1}\|$$

$$\|AX - I\| \leq (\|A\| \|A^{-1}\|) \|XA - I\|$$

$$\|AX - I\| \leq \kappa(A) \|XA - I\|, \text{ where } \kappa(A) = \|A\| \|A^{-1}\|$$

4. Let $A = \begin{pmatrix} \sqrt{2} & 0 & 1-\sqrt{2} \\ 0 & -\sqrt{3} & 0 \\ \sqrt{5} & 0 & -\sqrt{5} \end{pmatrix}$ and $b = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

a) Compute $\kappa(A)$ wrt $\|\cdot\|_\infty$

$$\|A\|_\infty = \max_{1 \leq i \leq n} \sum_{j=1}^n |a_{ij}| = 2\sqrt{5}$$

row 1: $|\sqrt{2}| + |0| + |1-\sqrt{2}| = \sqrt{2} + \sqrt{2} - 1 = 2\sqrt{2} - 1$

row 2: $|0| + |-\sqrt{3}| + |0| = \sqrt{3}$

row 3: $|\sqrt{5}| + |0| + |-\sqrt{5}| = 2\sqrt{5}$

$$\kappa(A) = \|A\|_\infty \cdot \|A^{-1}\|_\infty$$

$$= (2\sqrt{5}) \left(\frac{\sqrt{5} + \sqrt{2}}{\sqrt{5}} \right) = 2(\sqrt{5} + \sqrt{2})$$

$$A^{-1}: \det(A) = -\sqrt{3}(\sqrt{2}(-\sqrt{5}) - \sqrt{5}(1-\sqrt{2})) = \sqrt{15}$$

$$A^{-1} = \begin{pmatrix} 1 & 0 & \frac{1-\sqrt{2}}{\sqrt{5}} \\ 0 & -\frac{1}{\sqrt{3}} & 0 \\ 1 & 0 & -\frac{\sqrt{2}}{\sqrt{5}} \end{pmatrix}$$

$$\|A^{-1}\|_\infty = 1 + \frac{\sqrt{2}}{\sqrt{5}} = \frac{\sqrt{5} + \sqrt{2}}{\sqrt{5}}$$

row 1: $|1| + \left| \frac{1-\sqrt{2}}{\sqrt{5}} \right| = 1 + \frac{\sqrt{2}-1}{\sqrt{5}}$

row 2: $\left| -\frac{1}{\sqrt{3}} \right| = \frac{1}{\sqrt{3}}$

row 3: $|1| + \left| -\frac{\sqrt{2}}{\sqrt{5}} \right| = 1 + \frac{\sqrt{2}}{\sqrt{5}}$

b) $\tilde{A} = A + \delta A \quad \tilde{b} = b + \delta b$

Bound $\|\delta A\|_\infty$ and $\|\delta b\|_\infty$ in terms of $\varepsilon_{\text{mach}}$

$$|f(x) - x| \leq \varepsilon |x|$$

$$\|\delta A\|_\infty = \max_i \sum_j |\delta a_{ij}| \leq \max_i \sum_j \varepsilon_{\text{mach}} |a_{ij}| = \varepsilon_{\text{mach}} \max_i \sum_j |a_{ij}| = \varepsilon_{\text{mach}} \|A\|_\infty$$

$$\hookrightarrow |\tilde{a}_{ij} - a_{ij}| = |\delta a_{ij}| \leq \varepsilon_{\text{mach}} |a_{ij}|$$

$$\Rightarrow \|\delta A\|_\infty \leq \varepsilon_{\text{mach}} \|A\|_\infty$$

$$\|\delta b\|_\infty = \max_i |\delta b_i| \leq \max_i \varepsilon_{\text{mach}} |b_i| = \varepsilon_{\text{mach}} \max_i |b_i| = \varepsilon_{\text{mach}} \|b\|_\infty$$

$$\hookrightarrow |\tilde{b}_i - b_i| = |\delta b_i| \leq \varepsilon_{\text{mach}} |b_i|$$

$$\Rightarrow \|\delta b\|_\infty \leq \varepsilon_{\text{mach}} \|b\|_\infty$$

c) Exact: $Ax = b \quad \tilde{A} = A + \delta A \quad \tilde{b} = b + \delta b$

Rounded: $\tilde{A}\tilde{x} = \tilde{b}$

$$\tilde{A}\tilde{x} = (A + \delta A)\tilde{x} = b + \delta b \Rightarrow A\tilde{x} + \delta A\tilde{x} = b + \delta b \Rightarrow A\tilde{x} - Ax = \delta b - \delta A\tilde{x} \Rightarrow A(\tilde{x} - x) = \delta b - \delta A\tilde{x}$$

$$\Rightarrow \tilde{x} - x = A^{-1}(\delta b - \delta A\tilde{x}) \Rightarrow \|\tilde{x} - x\|_\infty \leq \|A^{-1}\|_\infty (\|\delta b\|_\infty + \|\delta A\|_\infty \|\tilde{x}\|_\infty) \quad \text{triangle inequality}$$

$$\Rightarrow \frac{\|x - \tilde{x}\|_\infty}{\|\tilde{x}\|_\infty} \leq \|A^{-1}\|_\infty \left(\frac{\|\delta b\|_\infty}{\|\tilde{x}\|_\infty} + \|\delta A\|_\infty \right) \Rightarrow \frac{\|x - \tilde{x}\|_\infty}{\|\tilde{x}\|_\infty} \leq \varepsilon_{\text{mach}} \underbrace{\|A\|_\infty \|A^{-1}\|_\infty}_{\kappa(A)} \left(\underbrace{\frac{\|b\|_\infty}{\|A\|_\infty \|\tilde{x}\|_\infty} + 1}_{\approx 2} \right) \quad \|b\| \approx \|A\tilde{x}\| \leq \|A\| \|x\|$$

$$\|\delta A\|_\infty \leq \varepsilon_{\text{mach}} \|A\|_\infty$$

$$\|\delta b\|_\infty \leq \varepsilon_{\text{mach}} \|b\|_\infty$$

$$\frac{\|x - \tilde{x}\|_{\infty}}{\|\tilde{x}\|_{\infty}} \leq 2 \cdot \kappa_{\infty}(A) \cdot \varepsilon_{mch} = \boxed{2\varepsilon_{mch}(2(\sqrt{5} + \sqrt{2}))}$$

5. a) in code below

b) in code below

c) from testing, it seems like running modified Gram-Schmidt twice seems to be the best method for reducing error. However, Householder QR is most likely the best since it has very minimal error and is considered more stable.

```

import numpy as np
from scipy.linalg import hilbert

def cgs(A):
    m, n = A.shape
    Q = np.zeros((m, n))
    R = np.zeros((n, n))
    for j in range(n):
        v = A[:, j].copy()
        for i in range(j):
            R[i, j] = np.dot(Q[:, i], A[:, j])
            v = v - R[i, j] * Q[:, i]
        R[j, j] = np.linalg.norm(v)
        Q[:, j] = v / R[j, j]
    return Q, R

def mgs(A):
    m, n = A.shape
    V = A.copy().astype(float)
    Q = np.zeros((m, n))
    R = np.zeros((n, n))
    for i in range(n):
        R[i, i] = np.linalg.norm(V[:, i])
        Q[:, i] = V[:, i] / R[i, i]
        for j in range(i + 1, n):
            R[i, j] = np.dot(Q[:, i], V[:, j])
            V[:, j] = V[:, j] - R[i, j] * Q[:, i]
    return Q, R

def householder_qr(A):
    m, n = A.shape
    R = A.copy().astype(float)
    Q = np.eye(m)
    for k in range(n):
        x = R[k:, k]
        e1 = np.zeros_like(x)
        e1[0] = 1.0
        # Householder vector v
        v = x + np.sign(x[0]) * np.linalg.norm(x) * e1
        v = v / np.linalg.norm(v)
        # Apply to R
        R[k:, k:] -= 2.0 * np.outer(v, np.dot(v, R[k:, k:]))
        # Accumulate Q
        H_k = np.eye(m)
        H_k[k:, k:] -= 2.0 * np.outer(v, v)
        Q = Q @ H_k
    return Q, R

# Testing with H7
n = 7
H7 = hilbert(n)

# (a) CGS and MGS
Q_cgs, _ = cgs(H7)
Q_mgs1, R_mgs1 = mgs(H7)

# (b) MGS twice
Q_mgs2, R_mgs2 = mgs(Q_mgs1)

# (c) Householder
Q_hh, R_hh = householder_qr(H7)

# Results
results = {
    "CGS": np.linalg.norm(Q_cgs.T @ Q_cgs - np.eye(n), ord=2),
    "MGS (1x)": np.linalg.norm(Q_mgs1.T @ Q_mgs1 - np.eye(n),
ord=2),
    "MGS (2x)": np.linalg.norm(Q_mgs2.T @ Q_mgs2 - np.eye(n),
ord=2),
    "Householder": np.linalg.norm(Q_hh.T @ Q_hh - np.eye(n),
ord=2)
}

for method, error in results.items():
    print(f"{method:15}: Error = {error:.2e}")

''' TERMINAL OUTPUT
CGS                : Error = 5.09e-02
MGS (1x)           : Error = 1.38e-09
MGS (2x)           : Error = 2.41e-16
Householder        : Error = 7.92e-16
'''

```